Year 13 Maths - Proof and mathematical communication

Similar questions - Algebraic proof and counter-examples

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Prove that $n^3 + 5n$ is divisible by 6 for every positive integer n. **[4]**

(b) A student claims that $n^3 + 5n$ is divisible by 12 for every positive integer n. Disprove this claim. **[1]**

2 A student claims that for every integer $m > 1$, the numbers $m^2 - 1$, $2m$ and $m^2 + 1$ form a Pythagorean triple.

(a) Prove that the claim is true. **[3]**

(b) Find the triple that has hypotenuse 65. **[2]**

(c) Prove that when m is odd, all three numbers in the triple are even. **[2]**

Question 1

(a) $n^3 + 5n = (n^3 - n) + 6n = (n-1)n(n+1) + 6n$. The product of three consecutive integers is divisible by $3! = 6$ (one factor is a multiple of 3 and at least one is even), and $6n$ is divisible by 6. So the sum is divisible by 6

(b) $n = 1$: $1 + 5 = 6$, which is not divisible by 12

Question 2

(a) $(m^2-1)^2 + (2m)^2 = m^4 - 2m^2 + 1 + 4m^2 = m^4 + 2m^2 + 1 = (m^2+1)^2$. All three numbers are positive integers when $m > 1$, so they form a Pythagorean triple

(b) $m^2 + 1 = 65$ gives $m = 8$. Triple $(63, 16, 65)$, check: $3969 + 256 = 4225$

(c) If m is odd then m^2 is odd, so $m^2 - 1$ and $m^2 + 1$ are even; $2m$ is always even